

Week Three Technical Report

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Teach2Give QA & QE Training Program

Overview

This week introduced the fundamentals of Microsoft SQL Server. The sessions covered data types, data definition, constraints, querying data, and different operations for managing and retrieving structured data in a relational database.

Training Modules Covered

We began with SQL Server data types. Then we moved to Data Definition Language, including creating and dropping databases, schemas, and tables, as well as altering and truncating them. We also looked at constraints like PRIMARY KEY, FOREIGN KEY, NOT NULL, and UNIQUE. Core querying followed with SELECT statements, aliases, filtering, and sorting with ORDER BY. The week continued with limiting rows, joining tables, subqueries, and grouping. We ended by practicing data modification.

Practical Exercise and Project

The exercises involved creating and managing databases, applying constraints, writing queries with conditions, sorting, and joins. We also practiced grouping data and performing transactions that simulated how SQL Server is applied in real database tasks.

Challenges Faced

The wide scope of topics covered this week was demanding, especially advanced grouping operations and joins. Extra practice was needed to understand these areas better and keep up with the pace of the training.

Recommendations

Breaking down topics into smaller study sessions each day makes it easier to understand complex concepts. Regular practice with exercises also helps in building confidence, especially when handling joins and grouping operations.

Conclusion

This week provided a solid introduction to SQL Server, from defining and managing data structures to querying and modifying records. Although the pace was intense, consistent practice made it possible to follow through. The experience emphasized the importance of patience and steady effort in learning technical topics.

Signature:

David Karanja